
Exercise Sheet 5:
Empirical distributions and learning

Foundations of Neuro-Symbolic AI
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Umbrella

Let us observe on 10 days states the variables

- Variable X_0 : It rains
- Variable X_1 : Alice brings an umbrella
- Variable X_2 : Bob brings an umbrella

The observations are:

Day	Rain (X_0)	Alice (X_1)	Bob (X_2)
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	0
8	1	1	1
9	1	1	1

Task 1:

Consider the empirical distribution of the variables X_0, X_1, X_2 , defined by this dataset.

- (i) Find a tensor network decomposition of the empirical distribution.
- (ii) Describe the contraction computing the rate of the days, on which Alice brings an umbrella exactly when Bob brings an umbrella. Simplify the contraction exploiting the directed notation and compute the rate.

Task 2:

Find the conditional independencies, which hold for the variables X_0, X_1, X_2 with respect to the empirical distribution.

Task 3:

Learn a Bayesian network by minimizing the unregularized Negative Log-Likelihood (NLL), given the empirical distribution.

- (i) Find the directed hypergraphs representing some of the conditional independencies with respect to the empirical distribution.
- (ii) Choose one of those hypergraphs and determine the conditional probability distributions that decorate the directed hypergraph.

Task 4:

Let us extend the dataset by day 10, on which we observe:

Day	Rain (X_0)	Alice (X_1)	Bob (X_2)
10	1	1	1

What would be the result of task 2 and 3, when using the extended dataset?